CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Tech. Electrical (Electrical Power System) Examination

December 2012

Subject: EE702 Modeling & Simulation of Electrical Machines

Date: 19.12.2012, Wednesday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Draw the generalized mathematical model of a polyphase induction machine. Write down the voltage equations for this model and obtain there from the equivalent circuit for polyphase induction motor. [10]

Q - 2 (a) Explain the various reference frames and discuss its applications in electrical machines analysis. Give the transformation matrix to transform parameters to stationary reference frames. [07]

OR

Q - 2 (a) Discuss various inductances of synchronous machines. And derive the equations of the same. [07]

Q - 2 (b) Discuss the General Identical transformations of any 3 phase (a, b, c) balanced rotating quantities to 2 phase (q, d). Obtain the transformation matrix, & inverse transformations. [08]

OR

Q - 2 (b) Draw and explain Free Running and Steady state characteristic of Induction Machine. [08]

Q - 3 Deduce Park's transformations relating the 3-phase currents of a synchronous machine to its corresponding d-q axes currents. Express 3-phase currents in terms of d-q axes currents and its inverse. Describe using physical concepts, how these transformations can be used for a polyphase induction motor. [10]

OR

Q - 3 (a) Express the voltage equations in the arbitrary reference frame for 2 phase resistive circuits. If 1.) Ra=Rb=Rs 2.) Ra not equal Rb. [07]

Where $Ks = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$.

(b) Discuss the effect of transformer voltage and speed voltage in electrical machines. [01]

(c) What do you mean by Stored energy and Co-energy in an electromagnetic system? [02]

SECTION – II

Q - 4 Explain the concept, construction and working of a brushless DC machines, along with its advantages and disadvantages of the machines [10]

Q - 5 (a) Derive the voltage equations for Kron's primitive machine in matrix form. What observations are made from the impedance matrix of this machine? [08]

Q - 5 (b) The energy stored in the coupling field of a magnetically linear system with two electrical inputs may be expressed as [08]

$$Wf(\lambda 1, \lambda 2, x) = 1/2\ B11\ \lambda 12 + B12\ \lambda 1\ \lambda 2 + 1/2\ B22\ \lambda 22$$

Express B11, B12 and B22 in terms of inductances L11, L12 and L22.

OR

Q - 5 (a) Derive the voltage equations for the generator and motor used in a interconnected system in WL method. [08]

Q - 5 (b) Carry out the transient analysis of a separately excited dc motor based on Kron's primitive machine. [08]

Q - 6 Derive the expression for a current operated system for an electromechanical energy conversion. [09]

OR

Q 6 (a) Derive the transformation matrix for transforming the variables from one reference frame to another. [04]

(b) Develop the block diagram for a dc motor connected with viscous load. [05]

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